

ENERGY

CREATING ELEMENTS

Stellar fusion is responsible for the variety of elements in the universe. Heavy elements are created as dying stars collapse, in the merging of neutron stars, or during other high-energy astrophysical events.

COMBINING NUCLEI

Two or more nuclei combine to form a larger nucleus in a process called nuclear fusion. When light nuclei fuse, they lose a little mass, which is released as energy. The repulsive Coulomb forces between positively charged nuclei make fusion impossible under all but the most extreme conditions, such as inside stars. Under these conditions, nuclei can tunnel (see p.71) through the Coulomb barrier and be brought close enough to fuse.

"I would like nuclear fusion to become a practical power source. It would provide an inexhaustible supply of energy, without pollution or global warming."

Stephen Hawking